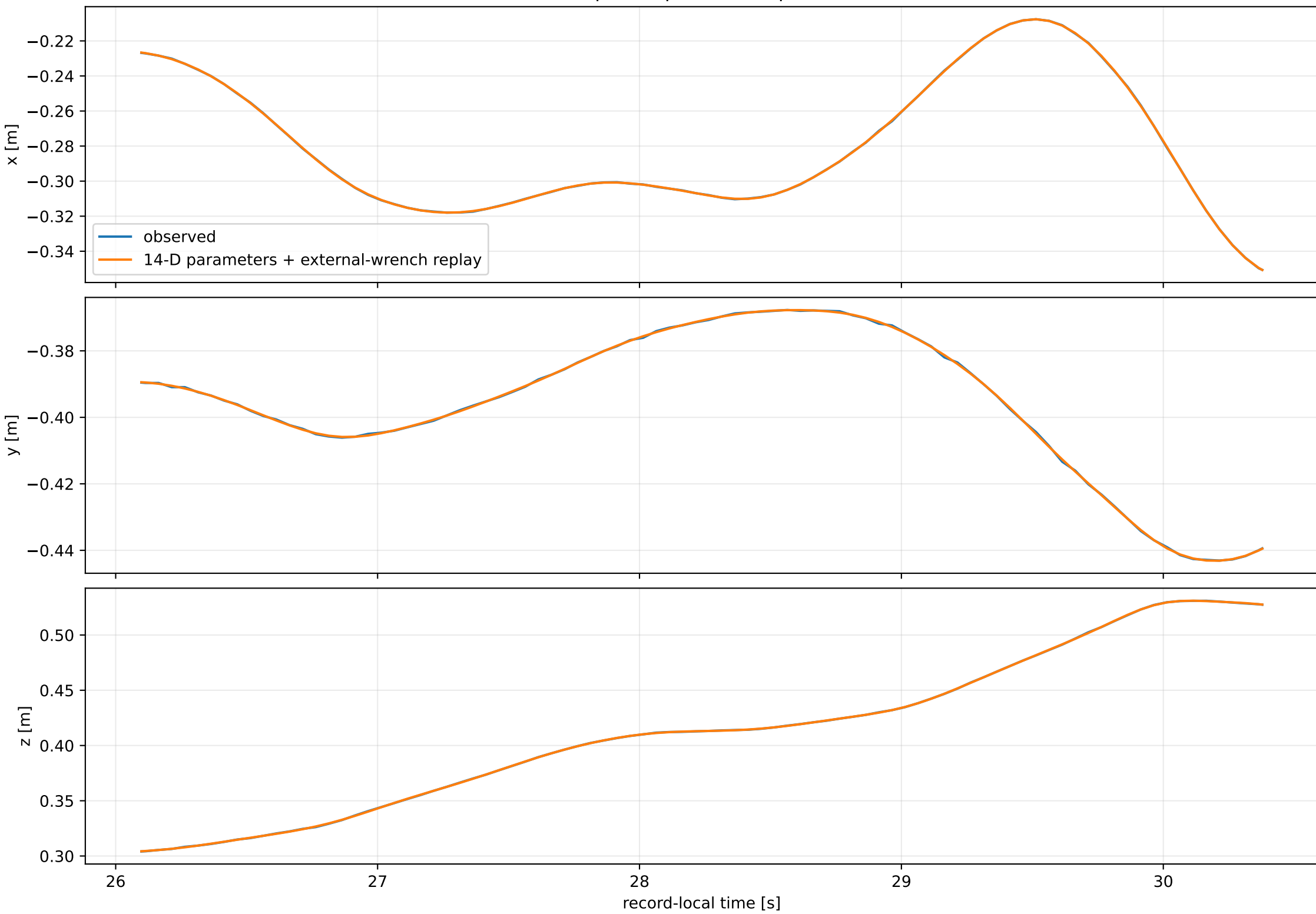
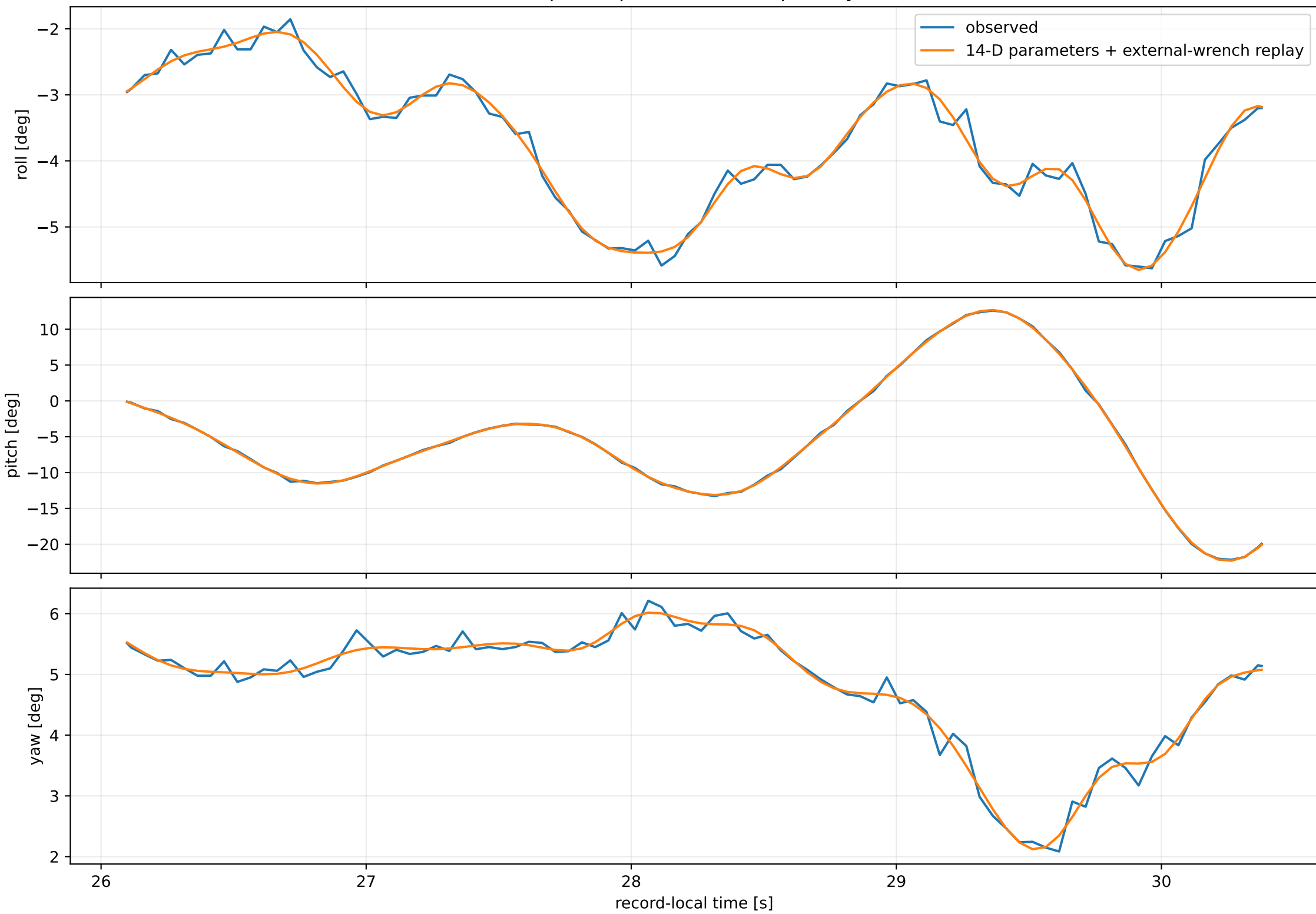


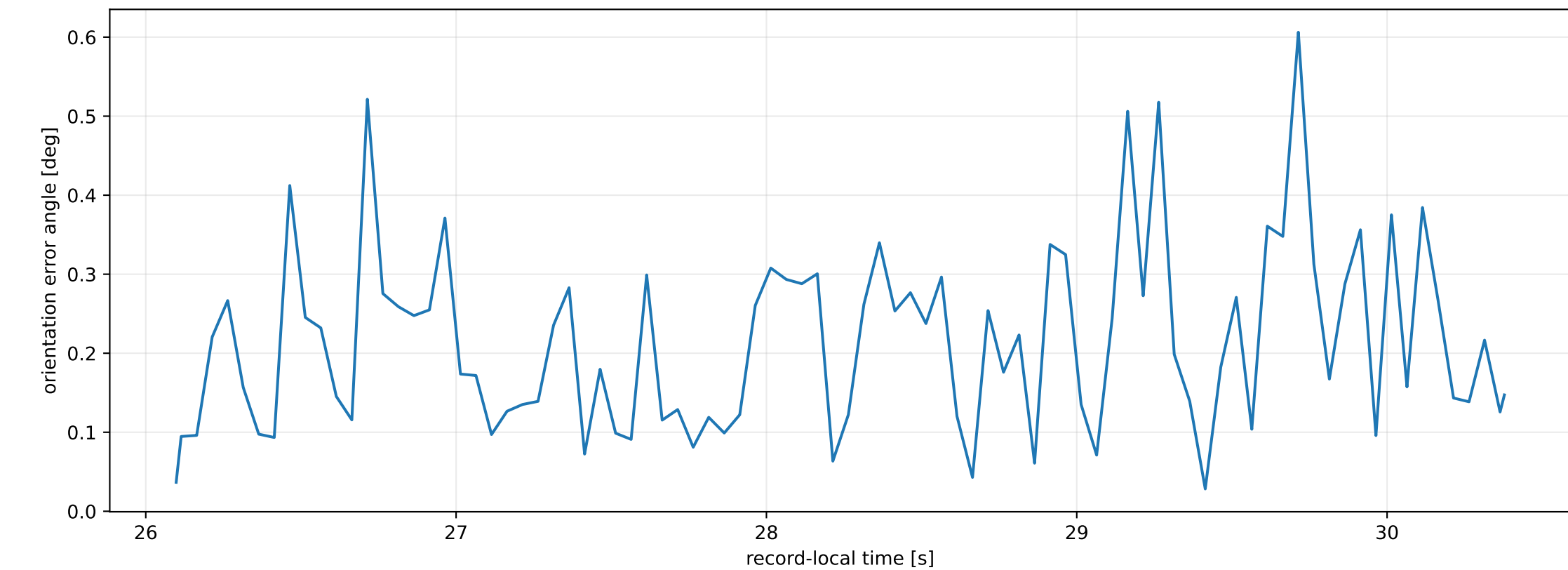
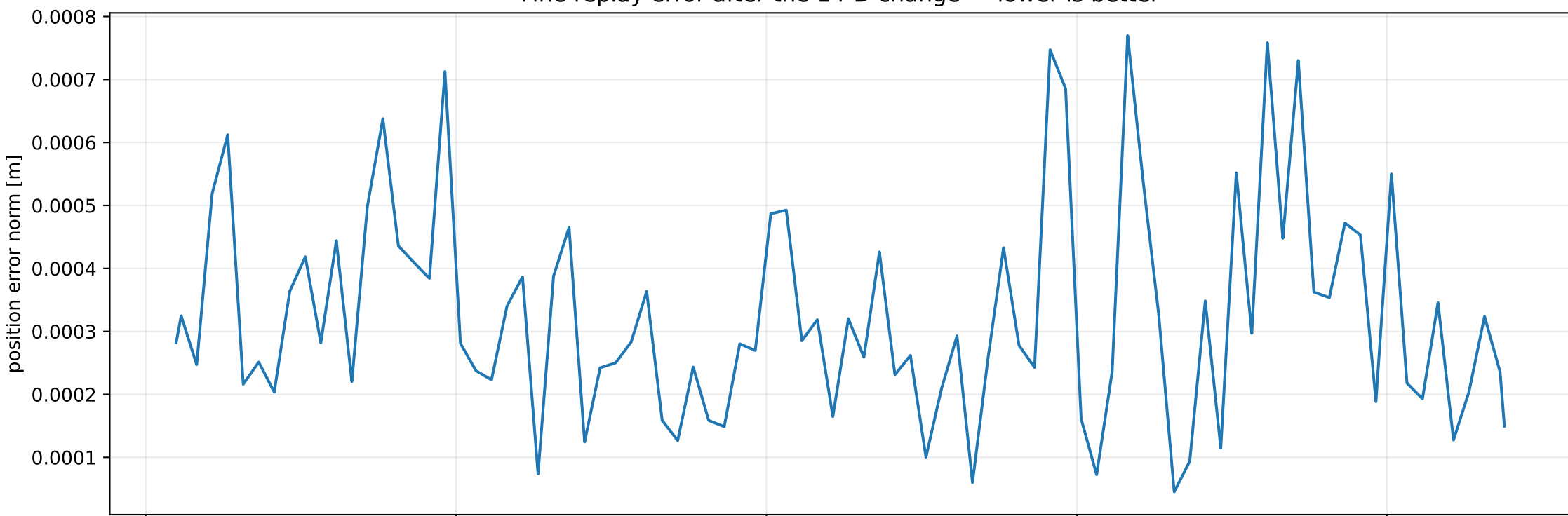
Fine pose reproduction: position



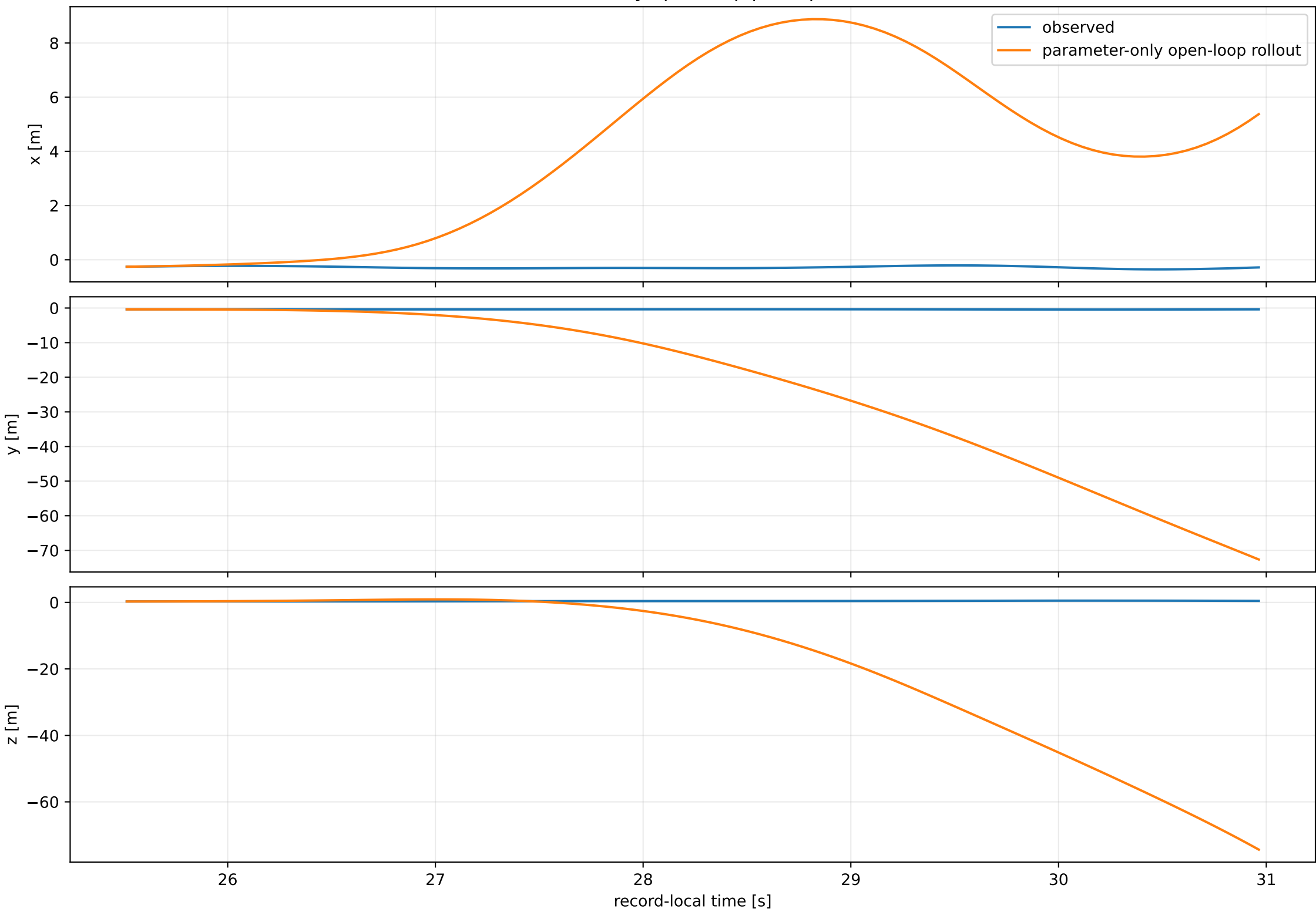
Fine pose reproduction: roll / pitch / yaw



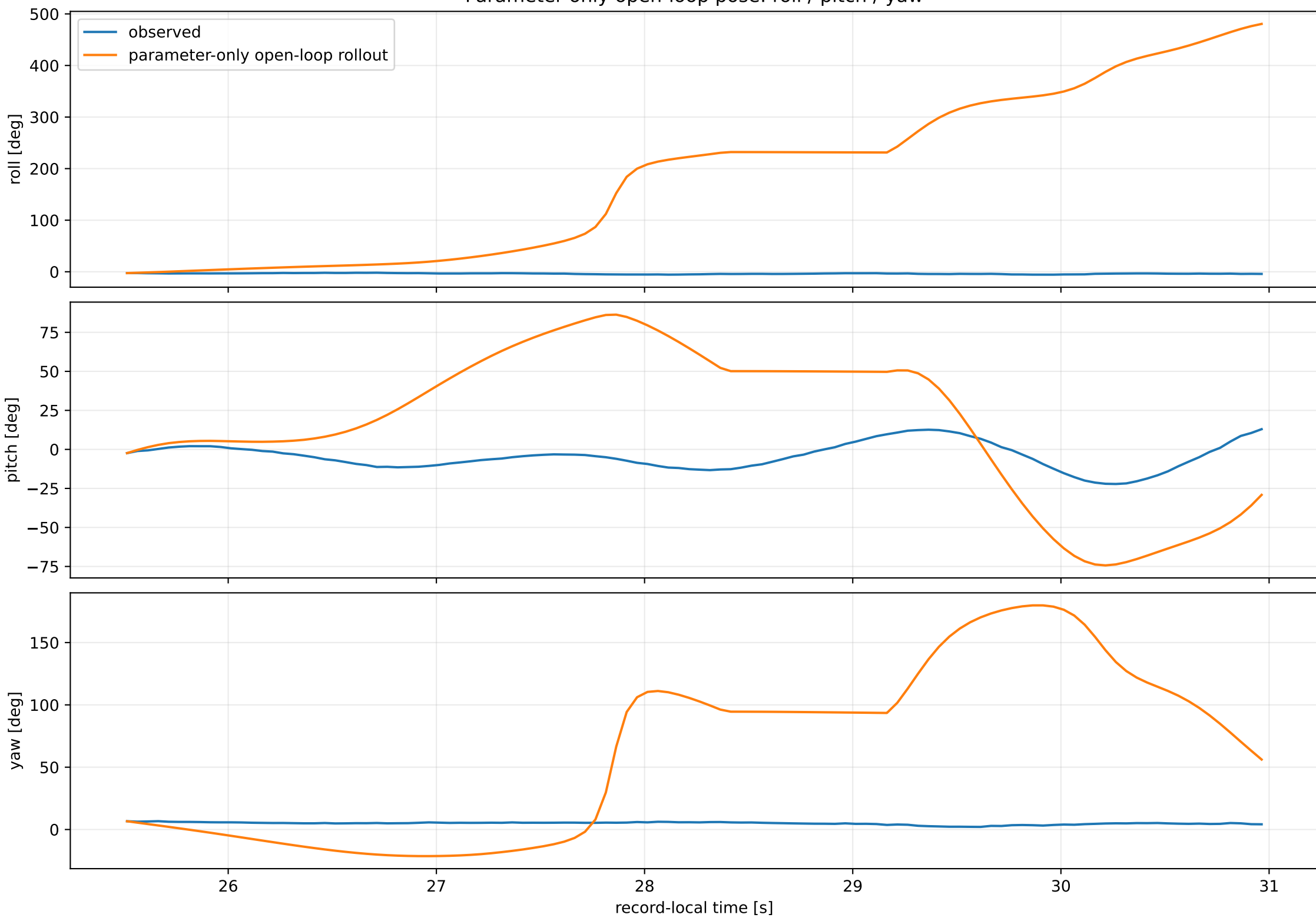
Fine replay error after the 14-D change — lower is better



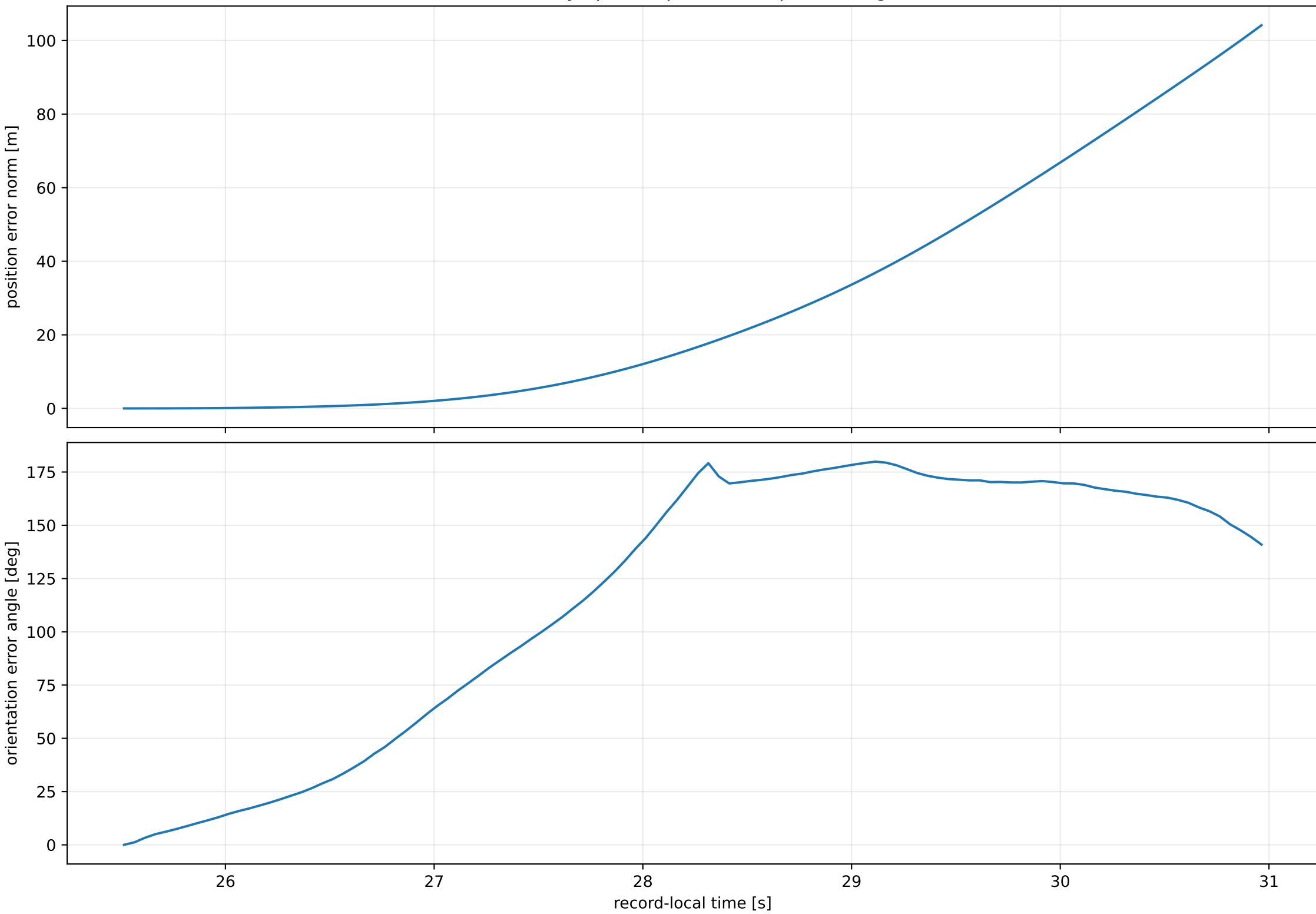
Parameter-only open-loop pose: position



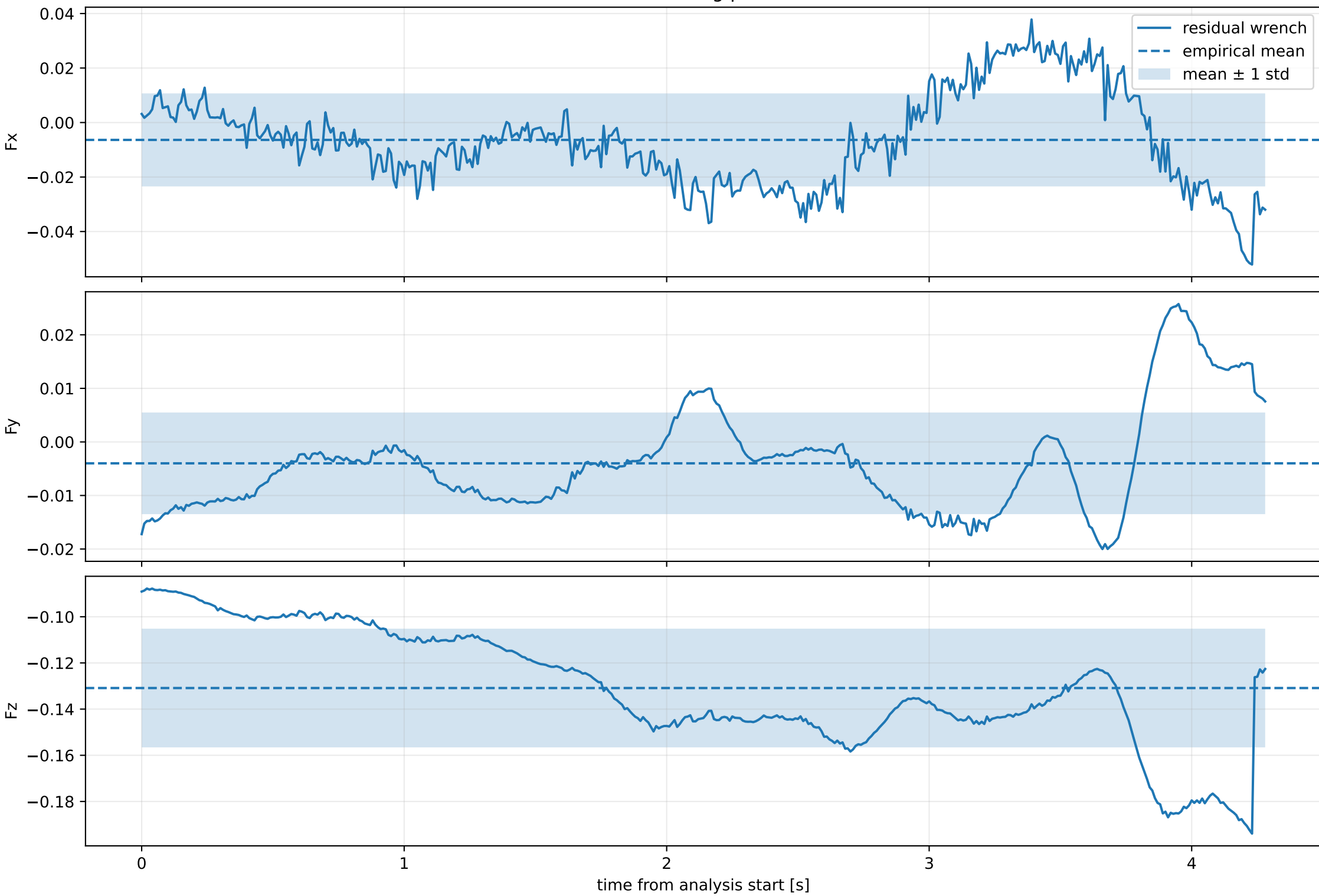
Parameter-only open-loop pose: roll / pitch / yaw



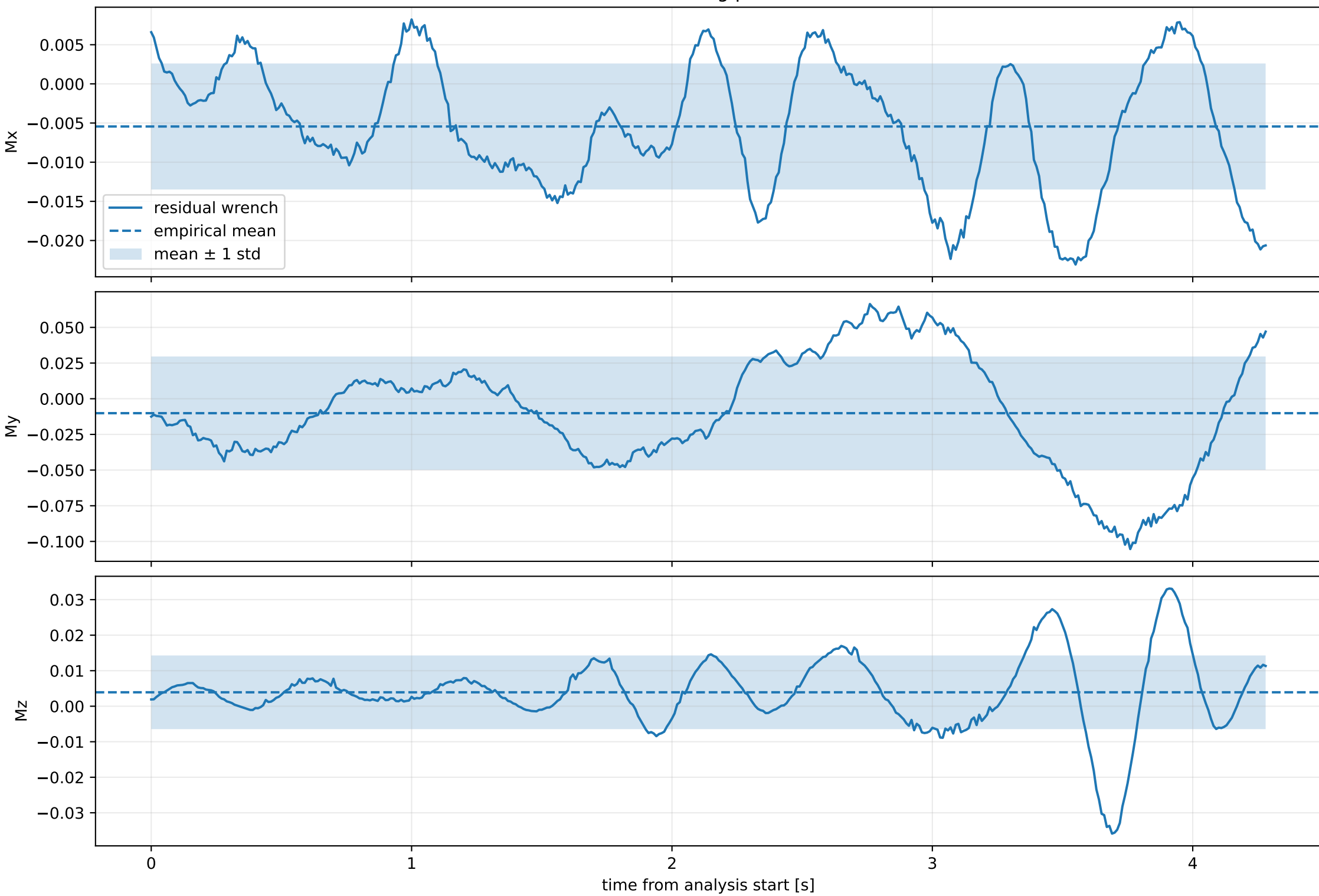
Parameter-only open-loop rollout (separate diagnostic)



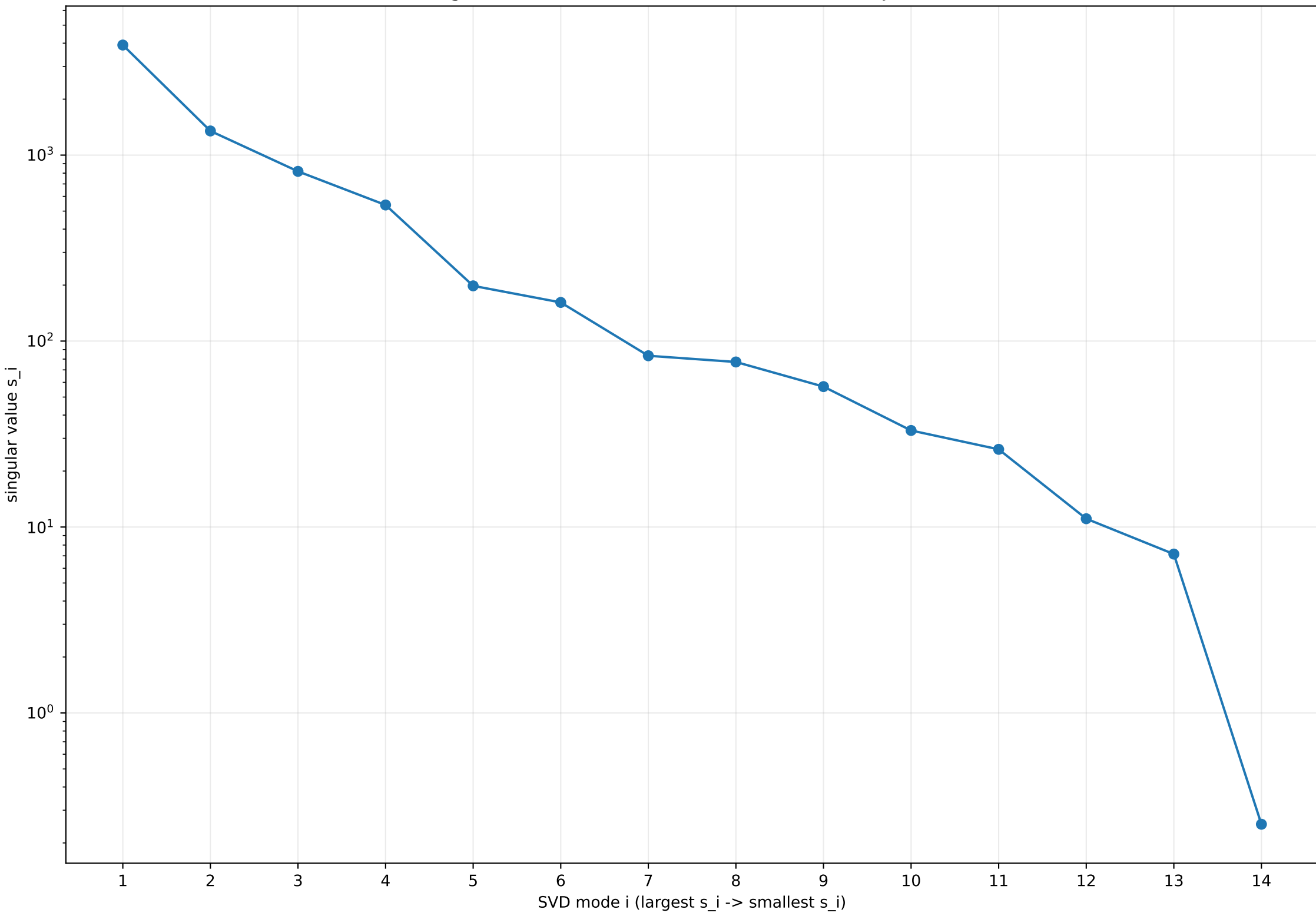
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



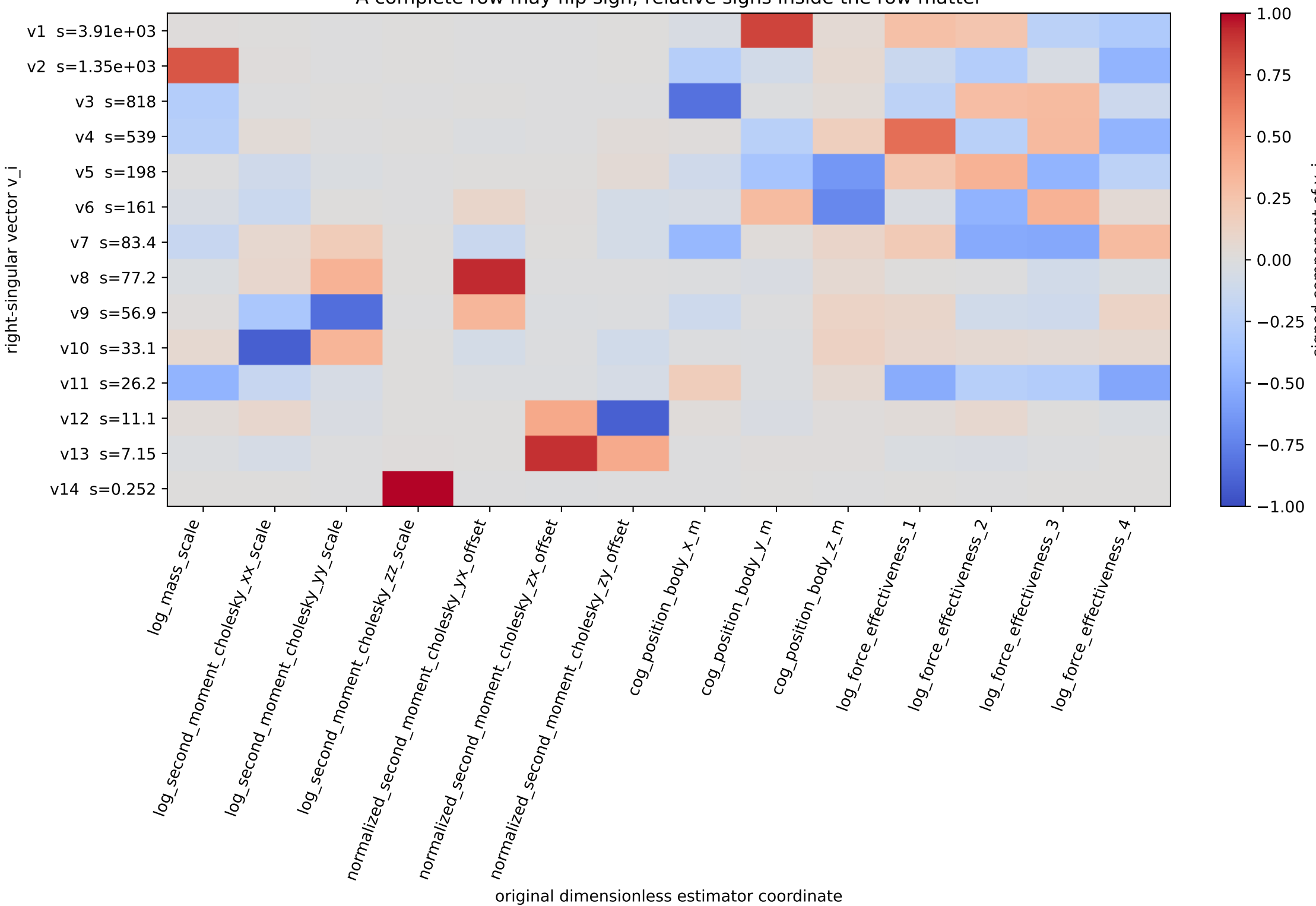
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36657792173

inertia J [kg m²]

[0.065002085 7.3035451e-06 1.8373057e-05]
[7.3035451e-06 0.06503052 9.7315286e-08]
[1.8373057e-05 9.7315286e-08 0.12899223]

principal moments [kg m²]

[0.065000314, 0.065032286, 0.12899223]

CoG position in body frame [m]

[-0.0027958637, 0.00039412092, 0.0095758676]

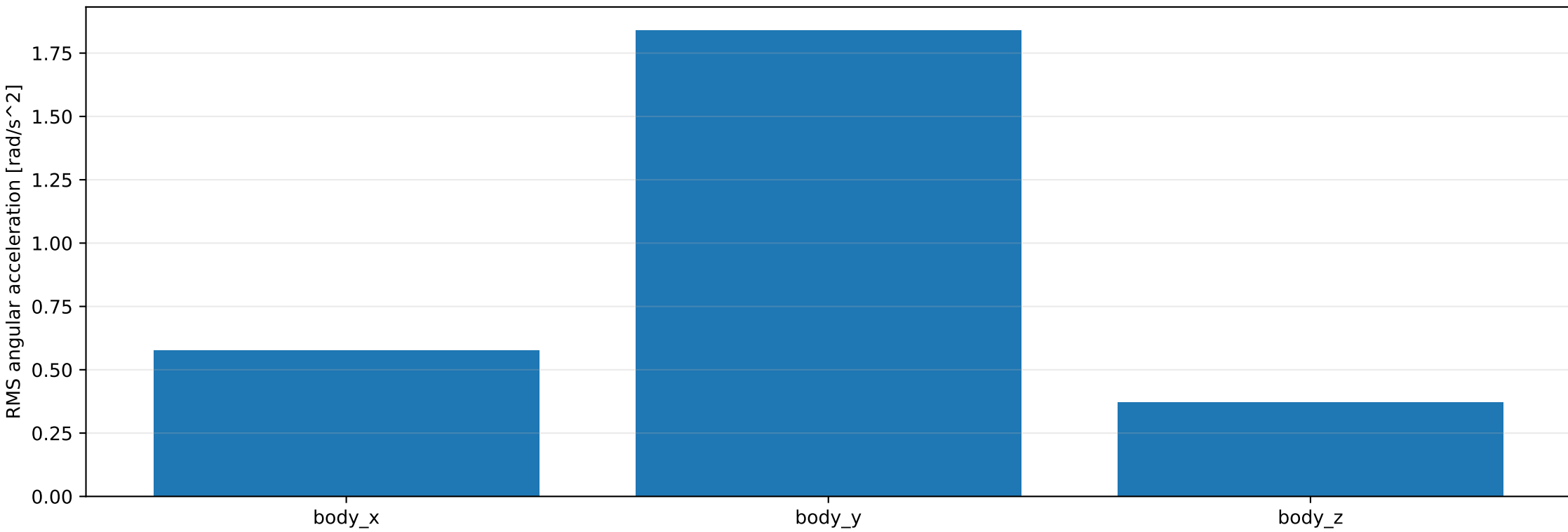
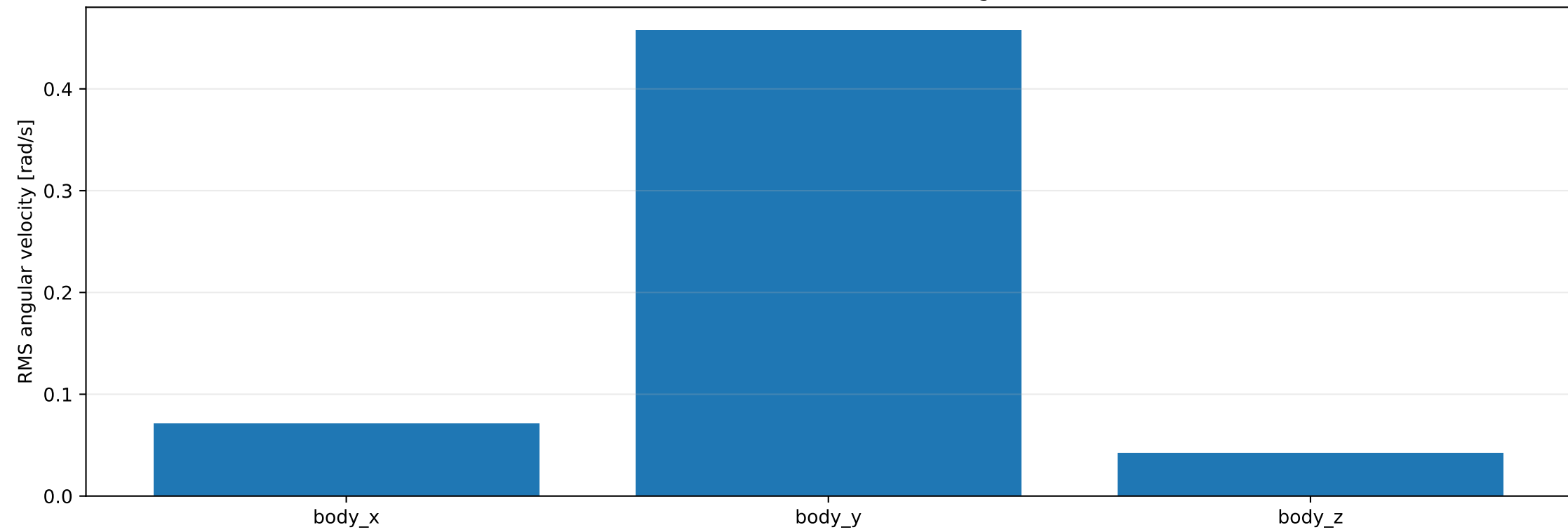
rotor force effectiveness

[0.95970893, 0.97113893, 0.9653399, 0.95416058]

selected delay [s]

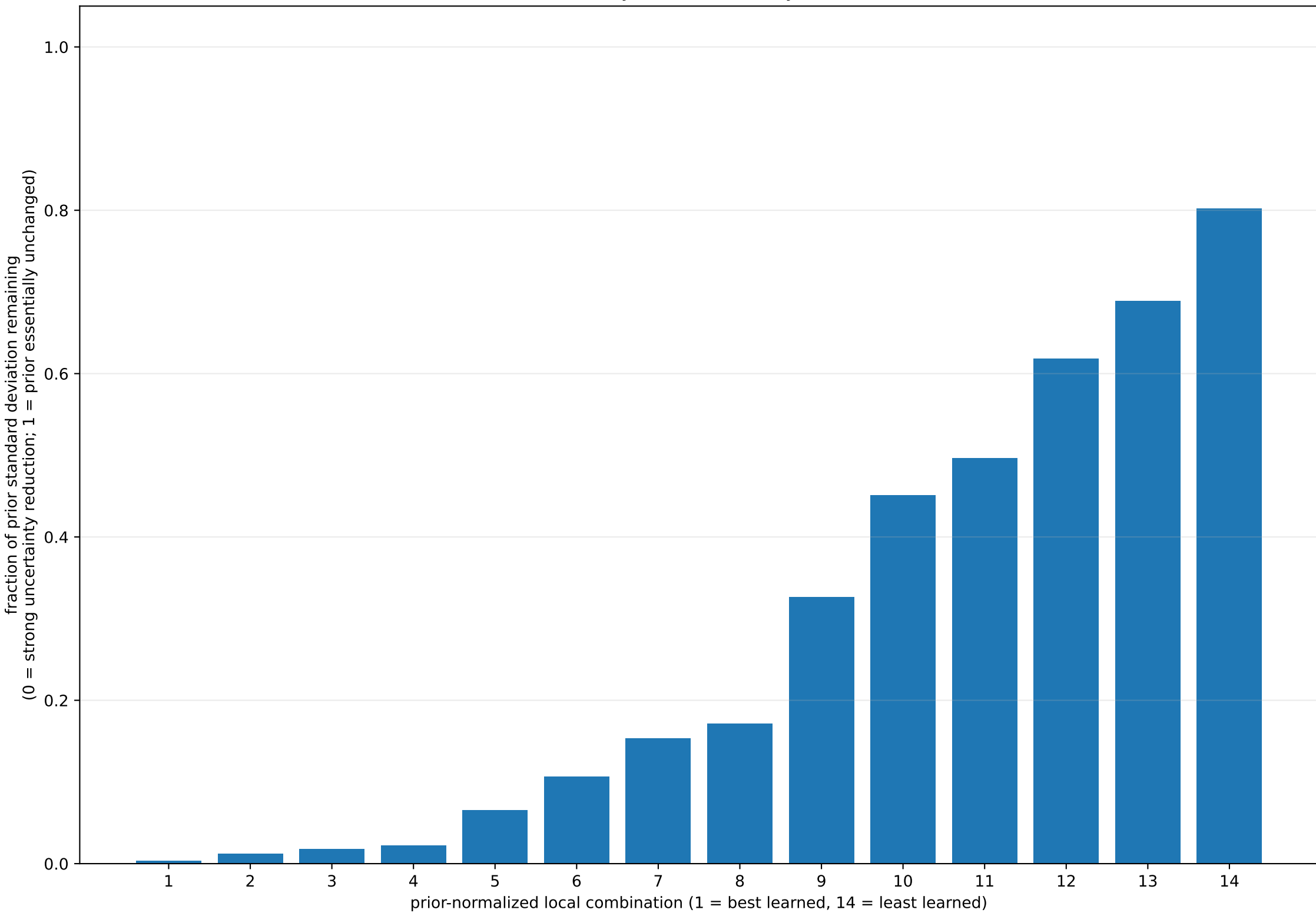
0.04769477421

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

